

W05 - Component Test Case Set

ScamGuard Component / Unit Testing Package

Team: ScamGuard - แอปตรวจสอบรูปภาพตัดต่อที่ถูกนำมาหลอกลวง

Course: ENGSE601 - Week 05 Output (W05-OUT)

SRS Baseline: 05_Software_Requirement_Specification.md v1.1 (September 12, 2026)

Code Baseline: Panuwat-ta/project branch main, commit 66bc9e4a

Critical Requirements: FR-ANALYSIS-02 | FR-ANALYSIS-04 | FR-AUTH-01 | FR-PDPA-01

Result: 18 cases - 7 Pass | 9 Fail | 1 Not Ready | 1 Needs Clarification

Date: 2026-09-22

1. Component Test Scope and Preparation (W05-01)

Field	Team record
Project/team	ScamGuard - Scam Image Detection Application
Requirement source/version	https://github.com/Panuwat-ta/project/tree/main/Document/srs , branch main, SRS v1.1 dated 2026-09-12
Source-code baseline	https://github.com/Panuwat-ta/project , branch main, commit 66bc9e4a
Execution date	2026-09-22
Selected requirements	FR-ANALYSIS-02, FR-ANALYSIS-04, FR-AUTH-01, FR-PDPA-01
Primary author / verification owner	ภานุวัฒน์
Assigned peer reviewer	เอกพันธ์ - reviewer decision/sign-off is still pending in repository evidence
Out of scope	Dataset-level Accuracy/F1/mDice, GPU performance, full System/UAT, stakeholder approval, unstated retroactive research-data behavior

Sources available

Artifact	Path / version	Status	How used
Canonical SRS	Document/srs/05_Software_Requirement_Specification.md v1.1	Existing	Acceptance basis for selected requirements
ENGSE212 source	Panuwat-ta/project main 66bc9e4a	Existing	Code/logic under verification
Existing unit test	server/tests/utils/test_risk_calculator.py	Existing	Executed risk-calculator evidence
Component probe	work/probes/component_probe.py	Existing	Reproducible risk/auth/route/visual probe
Raw evidence	work/evidence/E05-*.txt	Existing	Frozen baseline and execution output

Verification techniques used

- Example-based component tests for SRS risk-score examples
- Boundary-value checks around Low/Medium/High transitions
- Negative handler probe for mandatory System Consent
- Static source/schema/route inspection for explicit SRS contracts
- Fake in-memory DB and test-only environment; no production secret/database is used

2. Component Test Case Set (W05-02)

2.1 Visual Analysis - FR-ANALYSIS-02

ID	AC	Technique	Expected	Actual evidence	Result
CT-01	AC-1	Contract inspection	Separate forgery_confidence 0-100 traceable to forgery signal	No forgery_confidence; worker exposes visual_risk_score from SegFormer max probability	Fail
CT-02	AC-2	Contract inspection	Separate ai_gen_confidence from an AI-Gen detector	ai_gen_prob is the same SegFormer max probability; no separate AI-Gen output	Fail
CT-03	AC-3	Readiness check	Visual-score behavior can be tested from canonical independent signals	Required separate signal contracts are not available on main	Not Ready

FR-ANALYSIS-02 AC-5 (GPU inference <= 10 seconds) is deferred to NFR testing because Week 05 has no valid performance evidence.

2.2 Risk Calculation - FR-ANALYSIS-04

ID	AC	Input	Expected	Actual evidence	Result
CT-04	AC-1	50/85/0	90, High, primary visual	90 high visual True	Pass
CT-05	AC-2	100/100/100	cap at 100, High	100 high visual True	Pass
CT-06	AC-3	10/0/0	10, Low	10 low textual False	Pass
CT-07	AC-4	30/0/0	30, Low	30 low textual False	Pass
CT-08	AC-5	55/0/0	55, Medium	55 medium textual False	Pass
CT-09	AC-6	80/0/0	80, High	80 high textual False	Pass
CT-10	AC-7	stated total 65 / visual 85	High by Visual Override	Hybrid max uses visual in base, so total cannot be 65 when visual is 85	Needs Clarification

Supporting boundary probe: 39 -> Low, 40 -> Medium, 69 -> Medium, 70 -> High.

2.3 Registration - FR-AUTH-01

ID	AC	Technique	Expected	Actual evidence	Result
CT-11	AC-1	Handler probe + schema inspection	Response includes id, full_name, email, role, status, created_at; no password/hash	Keys are email, full_name, id, message, role; status and created_at absent	Fail
CT-12	AC-5	Negative handler probe: System Consent=false	HTTP 400 and no user/consent persistence	Handler returns UserResponse; fake DB receives User + ConsentLog(false,false)	Fail
CT-13	AC-1 subcondition	Handler probe: system=true, research=false	Optional Research Consent must not block registration	Registration succeeds and persists research=false	Pass

2.4 Consent Management - FR-PDPA-01

ID	AC	Technique	Expected	Actual evidence	Result
CT-14	AC-2	Model contract inspection	Event/log model supports consent_type, is_granted and separate records	Current model stores one row with two booleans	Fail
CT-15	AC-3	Route inventory	PUT /consent/research updates research consent and records update	Endpoint not present on main	Fail
CT-16	AC-4	Route contract inspection	GET /users/me returns current-user profile	Profile GET is /api/v1/auth/me; /api/v1/users/me is DELETE	Fail
CT-17	AC-5	Route inventory	GET /consent/logs returns consent history	Endpoint not present on main	Fail
CT-18	AC-3/5	Model contract inspection	Consent history/update supports updated_at	Current ConsentLog has created_at only	Fail

2.5 Result summary

Result	Count	Meaning
Pass	7	Verified behavior aligns with frozen Week 05 baseline
Fail	9	Current implementation does not align with baseline
Not Ready	1	Required component contract is unavailable for a valid acceptance test
Needs Clarification	1	Requirement example conflicts with baseline algorithm semantics
Total	18	Selected Week 05 component cases

3. Test Execution and Evidence (W05-03)

Evidence ID / file	Type	What it proves
E05-baseline.txt	Baseline record	ENGSE212 code commit, SRS version and SRS SHA-256
E05-pytest-component.txt	Executed unit test	test_risk_calculator.py: 1 passed in 0.03s
E05-component-probe.txt	Executed component probe	Risk examples/boundaries, auth behavior, route and visual contract inventory
component_probe.py	Reproducible probe source	Re-run logic with fake DB and test-only environment

Observed evidence highlights

- Risk examples: 50/85/0 -> 90 high ; 100/100/100 -> 100 high
- Grade boundaries: 39/40/69/70 -> Low/Medium/Medium/High
- system_consent=false still creates User + ConsentLog instead of HTTP 400
- Registration response lacks status and created_at
- No PUT /consent/research or GET /consent/logs route on frozen baseline
- Profile GET is /api/v1/auth/me ; /api/v1/users/me is DELETE
- Same SegFormer max probability feeds ai_gen_prob and visual_risk_score

Evidence integrity

Execution used an isolated origin/main archive. No .env , token, API key, production password or production database content is included. Static contract inspection is separated from runtime execution and is not presented as an end-to-end test. No unsupported model metric, GPU-time, stakeholder approval or peer-review approval is claimed.

4. Findings and Disposition (W05-04)

ID	Finding	Status	Owner	Next action
F01	Visual forgery/AI-Gen signals are not separated as required	Open	ກາບຸວັດນີ	Fix output contract/detector and re-test before System/UAT
F02	AC-7 example 65 total / 85 visual conflicts with Hybrid max+bonus	Needs Clarification	ກາບຸວັດນີ	Clarify SRS through controlled change; do not guess intent
F03	Registration response misses status and created_at	Open	ກາບຸວັດນີ	Fix response schema/handler and add contract test
F04	System Consent=false is accepted and persisted	Open	ກາບຸວັດນີ	Add guard before persistence and re-test negative case
F05	Consent persistence differs from event/log SRS contract	Open	ກາບຸວັດນີ	Reconcile schema/migration before integration testing
F06	Research-consent update/log endpoints are missing	Open	ກາບຸວັດນີ	Implement endpoints and component/API tests
F07	/users/me profile route differs between SRS and code	Open	ກາບຸວັດນີ	Select canonical route then synchronize SRS/code/test
F08	W03/W04 used SRS v1.0	Closed for W05	ກາບຸວັດນີ	Preserve as historical evidence; use v1.1 as current authority

5. Handoff to Week 06 and Later Testing

Destination	Week 05 handoff
Week 06 Integration Testing	F01, F05, F06, F07 are interface/data-contract risks; re-test after contracts stabilize
Week 07 System/UAT	F01, F03, F04, F06, F07 affect end-to-end journeys after fixes
Week 11 NFR Testing	GPU inference time and model-quality metrics need dedicated evidence

6. Peer Review Record

Field	Record
Artifact	Week 05 Component Test Case Set
Primary author	ภาณุวัฒน์
Assigned reviewer	เอกพันธ์
Author checklist	Baseline pinned; 18 cases traceable; raw evidence stored; findings have owner/next action; AI use declared
Reviewer date	Pending
Reviewer decision	Pending
Reviewer comments	Pending
Reviewer sign-off	Pending

Original v1 repository event: PR #5 merged at `ca354a8` , tag `w05-submission-v1` . GitHub showed no recorded independent PR review at verification time; this document does not infer a completed peer review.

7. AI Use Declaration

รายการ	บันทึก
Week / artifact	Week 05 / Component Test Case Set
ใช้ AI หรือไม่	Yes
เครื่องมือ/รุ่นเท่าที่ทราบ	ChatGPT GPT-5.6 Sol + Remote Desktop Commander
ใช้เพื่อ	อ่าน canonical SRS, ตรวจสอบ code/logic, จัด test cases, รับ probe, สรุป evidence/findings และจัดทำเอกสาร
Input ที่ให้ AI	Canonical Document / srs v1.1, ENGSE212 main 66bc9e4a, Week 05 guide และ repository evidence
การปกป้องข้อมูล	ไม่คัดลอก .env, token, API key, password หรือ production data ลง evidence
Accepted	SRS v1.1 เป็น authority; 18 cases; คง Fail/Not Ready ตามจริง; trace finding ไป evidence/next action
Modified	แยก static inspection จาก runtime evidence; W03/W04 เป็น historical handoff เท่านั้น
Rejected	การสร้าง threshold, model metric, stakeholder decision, Pass หรือ reviewer sign-off ที่ไม่มีหลักฐาน
วิธีตรวจสอบ	เทียบ expected กับ SRS; รับ pytest/probe บน isolated baseline; เก็บ raw output ใน work/evidence/
Verification owner/date	ภาณุวัฒน์ / 2026-09-22
คำยืนยัน	ไม่มีการสร้าง test result, screenshot, stakeholder decision, approval หรือ peer-review sign-off ที่ไม่ได้เกิดขึ้นจริง
แหล่งอ้างอิง	SRS: https://github.com/Panuwat-ta/project/tree/main/Document/srs ; Source: https://github.com/Panuwat-ta/project,branchmain